MIL-C-13753A(WC)
17 October 1972
SUPERSEDING
MIL-C-13753 (Ord)
5 November 1954

MILITARY SPECIFICATION
CHARGER, MANUAL, M6

1. SCOPE

1.1 This specification covers one type of manually operated charger for 20MM automatic guns, employed to effect the initial bolt movement necessary to start automatic firing or to restore automatic cycling after misfire.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-13931 - Cannons, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions

PUBLICATIONS

U. S. Army Weapons Command

D7227817, Sheet 1 - Charger, Manual, 20MM, M6: List of Drawings and List of Specifications
DL 7227817 - Index of Inspection Equipment Data Lists for Charger, Manual 20MM M6
PDS F7227817 - Packaging Data Sheet, Charger, Manual, 20MM M6

(Copies of specifications, standards, drawings, and publication, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First article. When specified in the contract, the contractor shall submit a first article. Unless otherwise specified (see 6.1.1). The first article shall be included in the pilot pack (see 5.1).

3.2 Materials and construction. This charger shall conform to this specification, MIL-C-13931, and the drawings and specifications listed in the List of Drawings and Specifications D7227817 Sheet 1.

3.3 Performance.

3.3.1 Charger operation. The assembled charger shall operate smoothly, without binding, through the complete rearward stroke and spring-actuated return of the slide, when operated as specified in 4.5.6.1. The protrusion and retraction of the latch at the termination of travel shall be completed without sticking or the application of undue force.

3.3.1.1 Operation of assemblies. Component assemblies containing moving parts shall be assembled in such a manner that these parts, when in motion, shall be free from abnormal resistance or restriction, and shall move without excessive looseness.

3.3.2 Load capacity. When operated as specified in 4.5.6.2 the charger shall withstand cable tension, applied through the handle, of 300 pounds in a direction parallel to the axis of the charger or, of 125 pounds in a direction perpendicular to the axis of the charger, without visible fracture in the cable, cable fittings, or attachment of jacket brackets, discernible slipping of cable fittings, distortion of the charger, or other damage apparent to the eye, or adversely affecting ease of operation.

3.3.2.1 Cable load. When tested as specified in 4.5.6.3 cables shall withstand the loads specified on the applicable drawings without fracture or slippage of wire or of fittings.

3.3.2.2 Jacket load. When tested as prescribed in 4.5.6.4 the jacket shall withstand a load of 800 pounds, applied in a direction parallel to the longitudinal axis without failure of the bracket joint.

3.4 Finish.

3.4.1 Painting. Chargers shall not be painted.

3.5 Workmanship. Workmanship shall be in accordance with MIL-C-13931. Chargers shall be completely assembled, free from discontinuities of coating, corrosion, foreign material, and such visible defects as may definitely reduce efficiency of operation or length of service.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions shall be as specified in MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.4 First article inspection. A first article shall be submitted for inspection in accordance with the contract requirements. The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspections specified herein and such other inspection as is necessary to determine compliance with requirements of the contract.

4.5 Quality conformance inspection.

4.5.1 Inspection sampling. Sampling of chargers and components thereof for non-destructive inspection shall be in accordance with MIL-STD-105, single sampling, except where 100 percent inspection or other procedure and sample size is specified herein or in the applicable documents.

4.5.2 Lot formation.

4.5.2.1 Assembled chargers. An inspection lot of assembled chargers shall consist of not more than 1000 chargers constructed exclusively from previously accepted parts and assemblies.

4.5.2.2 Components. Inspection lots of parts and assemblies shall be in accordance with MIL-C-13931.

4.5.3 Rejection. Unless otherwise specified, the rejection of lots of assemblies and component parts shall be in accordance with MIL-STD-105.

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4.5.3.1 Retests. At the discretion of the inspector, lots of chargers, or component parts, rejected for failure in mechanical test may be resubmitted for test after correction or replacement of the defective components. In the event of resubmission, the inspector shall inspect the resubmitted lot for all types or classes of defects that caused the lot to be rejected.

4.5.4 Classification of defects. Classification of defects and acceptable quality levels (AQL) for inspection in accordance with MIL-STD-105 shall be as follows:

4.5.4.1 Assembled charger.

Major defects, AQL: 0.40 percent defective:
Non-compliance with 3.3.2 in test specified by 4.5.6.2

4.5.4.2 Components.

4.5.4.2.1 Cradle mounting bracket.

Major Defects, AQL: 1.0 percent:
Width (0.748 - 0.005) over maximum or under minimum.
Location of three screw holes incorrect.
Screw holes below specified diameter.
Slot in forked leg below minimum width.
Counter-bores below specified diameter.

Minor defects, AQL: 2.5 percent:
Dimensional defects not classed as major defects.
Finish not as specified.

4.5.4.2.2 Guide bracket.

Major defects, AQL: 1.0 percent:
Incorrect distance, center to center, between screw holes.
Incorrect distance between screw hole and center of loop.
Undersize loop diameter or depth.

Minor defects, AQL: 2.5 percent:
Incorrect dimensions not classed as major defects.
Finish not as specified.

4.5.4.2.3 Cable, long.

Major defects, AQL: 1.0 percent:
Non-compliance with 3.3.2.1 in test specified by 4.5.6.3.
Oversize fitting.
Cable wires frayed at free end.
Excess projection through fitting.

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Minor defects, AQL: 2.5 percent:
Length not as specified.
Undersize fitting.

4.5.4.2.4 Cable, short.

Major defects, AQL: 1.0 percent:
Non-compliance with 3.3.2.1 in test specified by 4.5.6.3.
Oversize fitting.
Excess projection through fitting.
Length not as specified.

Minor defects, AQL: 2.5 percent
Undersize fitting.

4.5.4.2.5 Clamp, charger, assembly.

Major defects, AQL: 1.0 percent:
Interior dimensions of yoke (bores, slot, or pin hole) not as specified.
Diameters, threads, or axial dimensions of plunger not as specified.
Washer thickness not as specified.
Ring threads not as specified.
Clamp pin body diameter, body length to hole or hole diameter not as specified.
Spring, non-compliance with mandatory requirements on applicable drawing.
Dimension, end of plunger to axis of clamp pin (assembled clamp) not as specified.

Minor defects, AQL: 2.5 percent:
Dimensional defects not classed as major defects.
Surface roughness exceeding the specified degree.

4.5.4.2.6 Guide. (Component of catch assembly).

Major defects, AQL: 1.0 percent:
Overall length, width, or thickness, of body not as specified.
Location or size of pin hole in body not as specified.
Dimensions or location of radial slot in body not as specified.
Overall length or outside diameter of shaft not as specified.
Location or dimensions of slot or transverse notch in shaft not as specified.
Surface roughness exceeding the specified degree.

Minor defect, AQL: 2.5 percent:
Dimensional defects not classed as major defects.

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4.5.4.2.7 Housing, swivel.

Major defects, AQL: 1.0 percent:
Non-compliance with specified dimensions for locating or size of body pivot; bore, slot for pulley, pin hole, or annular groove.
Non-compliance with specified dimensions for axial length, width, or thickness, or bore of collar.
Position of collar on body not as specified.

Minor defects, AQL 2.5 percent.
Dimensional defects not classed as major defects
Surface roughness exceeding the specified degree.

4.5.4.2.8 Jacket.

Major defects, AQL: 1.0 percent:
Inside or outside diameter of tube not as specified.
Location of screw holes not as specified.
Screw holes below specified size.
Non-compliance of brazed joints with 3.3.2.2 when tested as specified in 4.5.6.4.

Minor defects, AQL: 2.5 percent
Dimensional defects not classed as major defects.
Surface roughness exceeding the specified degree.

4.5.4.2.9 Pin, locking (Component of pulley assembly).

Major defects, AQL: 1.0 percent:
Diameter of pin exceeding specified limits.
Length of pin not as specified.

Minor defects, AQL: 2.5 percent:
Dimensional defects not classed as major defects.
Surface roughness exceeding the specified degree.

4.5.4.2.10 Pin, pulley (Component of catch assembly).

Major defects, AQL: 1.0 percent:
Diameter of pin exceeding specified limits.
Length of pin not as specified.

Minor defects, AQL: 2.5 percent:
Dimensional defects not classed as major defects.
Surface roughness exceeding the specified degree.

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4.5.4.2.11 Pin, swivel (Component of pulley assembly).

Major defects, AQL: 1.0 percent:
Diameter of pin exceeding specified limits.
Length of pin not as specified.

Minor defects, AQL: 2.5 percent:
Dimensional defects not classed as major defects.
Surface roughness exceeding the specified degree.

4.5.4.2.12 Plug, front end.

Major defects, AQL: 1.0 percent:
Diameter of plug not as specified.
Hole locations not as specified.
Threads not as specified.
Hole diameter not as specified.

Minor defects, AQL: 2.5 percent:
Dimensional defects not classed as major defects.
Surface roughness exceeding the specified degree.

4.5.4.2.13 Plunger assembly.

Major defects, AQL: 1.0 percent:
Diameter of pulley not as specified.
Length and width of body not as specified.
Radius of pulley groove not as specified.
Width of slot in body not as specified.
Diameter of pin exceeding specified limits.
Length of pin not as specified.
Diameter and location of pulley hole not as specified.
Thickness of pulley not as specified.

Minor defects, AQL: 2.5 percent:
Dimensional defects not classed as major.
Surface roughness exceeding the specified degree.

4.5.4.2.14 Pulley, catch.

Major defects, AQL 1.0 percent:
Radius and depth of pulley groove not as specified.
Diameter and location of pulley hole not as specified.
Diameter of pulley not as specified.
Finish of pulley hole not as specified.

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Minor defects, AQL: 2.5 percent:
Dimensional defects not classed as major defects.
Surface roughness exceeding the specified degree.

4.5.4.2.15 Screw, guide bracket.

Major defects, AQL: 1.0 percent:
Diameter of head not as specified.

Minor defect, AQL: 2.5 percent:
Dimensional defects not classed as major defects.

4.5.4.2.16 Screw, Jacket.

Major defects, AQL: 1.0 percent:
Diameter of head not as specified.

Minor defects, AQL: 2.5 percent:
Dimensional defects not classed as major defects.

4.5.4.2.17 Sheave, swivel (Component of pulley assembly).

Major defect, AQL: 1.0 percent:
Diameter and location of pulley hole not as specified.
Radius and depth of pulley groove not as specified.
Diameter of pulley not as specified.

Minor defects, AQL: 2.5 percent:
Dimensional defects not classed as major defects.
Surface roughness exceeding the specified degree.

4.5.4.2.18 Body (Component of slide assembly.)

Major defects, AQL: 1.0 percent:
Width and depth of slot in body not as specified.
Hole locations not as specified.
Diameter of holes not as specified.
Length and width of body not as specified.
Curvature of body surface not as specified.

Minor defects AQL; 2.5 percent:
Dimensional defects not classed as major defects.
Surface roughness exceeding the specified degree.

4.5.4.2.19 Plunger (Component of slide assembly.)

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Major defects, AQL; 1.0 percent:
Length of plunger not as specified.
Radius of plunger head not as specified.
Diameter of plunger not as specified.
Finish on cylindrical surface of plunger not as specified.

Minor defects, AQL; 2.5 percent:
Dimensional defects not classed as major defects.

4.5.4.2.20 Pin, swivel (Component of slide assembly).

Major defects, AQL; 1.0 percent:
Diameter of pin exceeding specified limits.
Length of pin not as specified.

Minor defects, AQL; 2.5 percent:
Dimensional defects not classed as major defects.
Surface roughness exceeding the specified degree.

4.5.4.2.21 Latch (Component of slide assembly).

Major defects, AQL; 1.0 percent:
Location and diameter of pin hole not as specified.

Minor defects, AQL; 2.5 percent:
Dimensional defects not classed as major defects.
Surface roughness exceeding the specified degree.

4.5.4.2.22 Spring, return.

Major defects, AQL; 1.0 percent:
Non-compliance with mandatory requirements specified on the drawings

Minor defects, AQL; 2.5 percent:
Dimensional defects not classed as major defects.

4.5.4.2.23 Tube, return spring.

Major defects, AQL; 1.0 percent:
Inside diameter of tube not as specified.
Length of tube not as specified.
Finish on inside surface of tube not as specified.
Outside diameter of tube greater than specified.

Minor defects, AQL; 2.5 percent:
Dimensional defects not classed as major defects.
Surface roughness exceeding the specified degree.

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4.5.5 Examinations.

4.5.5.1 Inspection provisions. The chargers shall be inspected for conformance to requirements of the provisions of this specification, MIL-C-13931, and SQAP-APPENDIX-WVT.

4.5.5.2 Finish.

4.5.5.2.1 Painting. Each charger shall be visually inspected for compliance with 3.4.1. Failure to comply shall be cause for rejection.

4.5.5.3 Workmanship. Each charger shall be visually inspected for compliance with 3.5. Failure to comply shall be cause for rejection.

4.5.5.4 Final examination. Final examination of each charger shall be completed after testing and prior to preservation and packaging to determine compliance with this specification, MIL-C-13931, and as specified in the contract.

4.5.5.5 Inspection of preparation for delivery.

4.5.5.5.1 Examination. The classification of defects and the AQLs associated with individual defects for examination of the preparation for delivery are as follows:

Category	Defect	AQL	Inspection Method
Major			
101	Illegible or incorrect marking	1.0	Visual
102	Improper level of preservation, packaging or packing	1.0	Visual
103	Inadequate cleaning or drying	1.5	Visual
104	Improper preservative application	1.5	Visual
105	Missing or improper cushioning	1.5	Visual
106	Improper blocking and bracing	1.5	Visual/Manual
107	Improper closure or strapping of containers	1.5	Visual/Manual
Minor			
201	Workmanship	4.0	Visual

4.5.5.6 Inspection approval stamp. The application of the inspection approval stamp shall be specified in MIL-C-13931.

4.5.6 Tests.

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4.5.6.1 Manipulation. Each assembled charger and each component assembly containing moving parts shall be operated manually through the normal motions to determine compliance with 3.3.1 and 3.3.1.1, as applicable. For manipulation, the charger shall be attached, by means of the charger clamp and cradle mounting bracket, to a rigidly supported metal plate 0.365 - 0.010 inch thick. The plate shall be notched and drilled to fit the jacket bracket and charger clamp, and tapped for the cradle mounting bracket cap screw. The charger shall be operated through its complete cycle not less than 3 times, by pulling and releasing the handle. The directions of pull shall be approximately at right angles to the charger, successively upward, downward, and approximately horizontal to the axis of the charger, utilizing the extremes and midway positions respectively, of the range of rotation of the swivel housing. Any evidence of sticking, application of undue force, abnormal resistance, restriction, or excessive looseness shall be cause for rejection.

4.5.6.2 Load capacity. The chargers to be tested shall be attached to a supporting fixture as specified in 4.5.6.1 and a retaining stop shall be placed against the protruding edge of the slide body to prevent the resulting slide motion as tension is placed on the cable. The cable shall then be pulled straight back along the axis of the charger, with a force of 300 pounds, applied through the handle, after which the test shall be repeated utilizing a force of 125 pounds, applied in a direction approximately at right angles to the axis of the charger. At the conclusion of the test, the charger shall be examined for compliance with 3.3.2. Failure to comply shall be cause for rejection.

4.5.6.3 Cable load. A metal plate, containing a cap-shaped recess, formed to contain the fittings and having a hole through which to pass the cables being tested, shall be suspended horizontally by a suitable supporting fixture at such a height that the end of the cable will hang freely. A weight of 300 pounds shall be attached to the free end of the cable and, the cable, while under load, shall be examined for compliance with 3.3.2.1. Failure to comply shall be cause for rejection.

4.5.6.4 Jacket load. The jacket to be tested shall be inserted into a cylindrical steel tube of such length and diameter that the jacket will not pass completely into the tube but will be retarded by the protruding brackets resting on the edge of the tube. A snugly fitting cylindrical steel bar shall be inserted into the tube of the jacket to prevent buckling of the jacket when pressure is applied as specified herein. A small steel plate approximately 1 inch x 1/2 inch x 1/4 inch shall be placed between the bracket joint being tested and the edge of the cylindrical tube in such a way that the bracket will rest on the plate, approximately 1/2 inch above the edge of the tube. A force of 800 pounds shall be applied to the protruding end of the jacket, after which the test shall be repeated for the second bracket joint. At the conclusion of the tests, the brazed joints shall be examined for compliance with 3.3.2.2 Failure to comply shall be cause for rejection.

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4.5.7 Inspection equipment. Data lists containing equipment used in the inspection of parts, subassemblies, and assemblies are listed in DL 7227817 for chargers.

4.5.7.1 Acquisition, maintenance, and disposition. Unless otherwise specified (see 6.1.1), responsibility for acquisition, maintenance, and disposition of acceptance inspection and test equipment prescribed in the List of Inspection Equipment DL 7227817 for chargers, and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.7.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. On each contract, a pilot pack (see 6.4) shall be forwarded in accordance with 3.1. Pilot packs for the charger shall be cleaned, preserved, and packaged to the level of packaging, specified in the contract, and packed level C in accordance with the requirements of Packaging Data Sheet P7227817.

5.2 Preservation, packaging, and marking. The charger shall be unit packaged, packed, and marked in accordance with the requirements of Packaging Data Sheet P7227817.

5.3 Repair parts. Repair parts shall be unit packaged, packed, and marked in accordance with the requirements of the applicable packaging data sheet (see 6.1.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Requirements for submission of and shipping instructions for first article (see 3.1 and 4.4) and pilot pack (see 3.1 and 5.1).

(1) The contract should designate the examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

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(2) The contract should specify the sample size (i.e., the number of units) of the first article sample.

(3) The contract should designate the inspection and test data that is required to be furnished the Government when the contractor is required to perform either or both examinations and tests.

(4) The contract should designate at whose expense a retest may be performed.

(5) The contract should specify that the acceptable first article sample shall be delivered in accordance with the terms of the contract, or

(6) Any additional or extended examinations and tests beyond the scope of this specification shall be specified and identified as a separate item of the contract

c. Serial numbers for the chargers.

d. List of inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.5.7).

e. Extent of supplier's responsibility for Government-furnished equipment.

f. Selection of applicable levels of preservation, packaging, and packing (see 5.2)

g. Packaging instructions for repair parts (see 5.3).

6.1.2 Contract data requirements. Monthly reports of the results of final examination and performance testing shall be specified for delivery on DD Form 1423 in the contract.

6.2 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control Systems Summary List, DD Form 1660.

6.3 Unless otherwise specified, the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

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6.4 Definition of pilot pack. A requirement in conjunction with a specification or contract for a first article in order to base approval of packages.

Custodian:
Army—WC

Preparing activity:
Army—WC

Project No. 1005 A460

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